

CRF: $n = d_s + n_m$

The Powerset Origin of Matter and the Space–Matter Split

Ougit Jarittum

April 2026 ORCID: [0009-0009-4451-6811](https://orcid.org/0009-0009-4451-6811) CC-BY-NC 4.0

Companion to CRF ThreePhase v2 and CRF SO3 Proof. Discovered through pattern
mining of the Psi-cluster structure.

Abstract. The $n = 8$ modes of CRF’s Possibility Space split exactly as

$$n = d_s + n_m \quad 8 = 3 + 5$$

where $d_s = 3$ modes crystallise into spatial dimensions and $n_m = n - d_s = 2^{d_s} - d_s = 5$ modes become matter archetypes.

This split has a combinatorial origin: $n = 2^{d_s}$ counts all subsets of a d_s -dimensional axis set $\{x, y, z\}$. The $d_s = 3$ *singleton* subsets ($\{x\}, \{y\}, \{z\}$) become the three spatial directions. The remaining $n_m = 5$ *non-singleton* subsets ($\{\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x, y, z\}$) become the five matter archetypes.

Under $SO(3)$, these five decompose as: $\{\}$ = scalar (singlet), $\{x, y\}, \{x, z\}, \{y, z\}$ = vector (triplet), $\{x, y, z\}$ = pseudoscalar (singlet) — giving $1 + 3 + 1 = 5$.

Three consequences: (1) $n_m = 5$ is derived from $d_s = 3$ alone, confirmed by V18 (5 Psi-clusters). (2) The Bethe-to-CRF loop correction = $1/((n-1)n_m) = 1/35 = 2.857\%$ (actual 2.904%, gap 1.6%). (3) $\mathcal{G}$ is approximately universal across space and matter sectors ($\mathcal{G}_m/\mathcal{G} = D_0/D_{0,m} \approx 1.037, 3.7\%$).

The Split

$n = d_s + n_m$ — **exact**

The $n = 8$ modes of $\mathcal{P}$ -space split without remainder:

$$n = 2^{d_s} = d_s + \underbrace{(2^{d_s} - d_s)}_{n_m} \quad 8 = 3 + 5$$

d_s modes crystallise into spatial dimensions (via $SO(3)$ Proof).

$n_m = n - d_s = 5$ modes become matter archetypes (Psi-cluster attractor set).

Both are exact integers. No simulation or free parameter is involved.

The general formula $n_{\text{m}} = 2^{d_s} - d_s$:

d_s	$n = 2^{d_s}$	$n_{\text{m}} = n - d_s$	Space + Matter
1	2	1	1 + 1
2	4	2	2 + 2
3	8	5	3 + 5 (<i>our universe</i>)
4	16	12	4 + 12
5	32	27	5 + 27

The Powerset Origin

Powerset structure

The $n = 2^{d_s} = 8$ modes are the *powerset* of the d_s -dimensional axis set $\mathcal{A} = \{x, y, z\}$:

$$\mathcal{P}(\mathcal{A}) = \{ \emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x, y, z\} \}$$

Singleton subsets $\{x\}, \{y\}, \{z\}$ (size 1): these are the three independent directions — they become the $d_s = 3$ spatial dimensions.

Non-singleton subsets (size $\neq 1$): these are the $n_{\text{m}} = 5$ remaining degrees of freedom — they become the matter archetypes.

The split $n = d_s + n_{\text{m}}$ is the split of a powerset into singletons and non-singletons. It requires no further axiom: it follows from δ creating d_s independent distinctions, whose powerset is $n = 2^{d_s}$.

SO(3) Decomposition of the 5 Matter Modes

The five non-singleton subsets decompose into SO(3) representations:

Subset	SO(3) rep	J	Physical archetype
$\emptyset$	singlet	0^+	scalar matter
$\{x, y\}, \{x, z\}, \{y, z\}$	triplet	1	vector matter (3 types)
$\{x, y, z\}$	singlet	0^-	pseudoscalar matter
Total: 5	$1 + 3 + 1$		

The decomposition is:

$$n_m = \underbrace{1}_{\text{scalar}} + \underbrace{3}_{\text{vector}} + \underbrace{1}_{\text{pseudoscalar}} = 5$$

This is the representation content of the powerset complement under SO(3) — not assumed, but the natural consequence of $d_s = 3$ making SO(3) the symmetry group.

Confirmation: V18 simulation (N=500, 400 sweeps) detects exactly 5 peaks in the Ψ -histogram (Psi-speciation). The Ψ field encodes internal (non-spatial) mode identity — so 5 Psi-clusters = 5 matter archetypes, as derived.

The Triangular-Number Ratio

Both sectors have triangular-number structures:

$$n(n+1) = 72 = 2T_8 \quad (\text{space, from K35}) \quad n_m(n_m+1) = 30 = 2T_5 \quad (\text{matter})$$

Their ratio is the exact rational:

$$\frac{n(n+1)}{n_m(n_m+1)} = \frac{72}{30} = \frac{8 \times 9}{5 \times 6} = \frac{n}{n_m} \cdot \frac{n+1}{n_m+1} = \frac{8}{5} \cdot \frac{9}{6} = \frac{8}{5} \cdot \frac{3}{2} = \frac{12}{5} = 2.4$$

The factor $8/5 \approx \varphi$ (golden ratio, 1.1% gap) and $3/2$ (exact) appear together. This ratio is an exact consequence of the split $n = 8, n_m = 5$.

The Bethe Loop Correction Explained

The percolation threshold for 3D space locking (CRF ThetaTower v2):

$$f_{\text{II,crit}} = 1 - e^{-\sqrt{\varphi}/n} = 0.14700$$

The Bethe mean-field baseline $p_c = 1/(n-1) = 1/7 = 0.14286$ underestimates $f_{\text{II,crit}}$ by 2.904% due to loop corrections.

The loops arise from Ψ -clustering: the n_m matter-type clusters create triangles in the weight graph W_{ij} , raising the percolation threshold.

Loop correction from matter modes — gap 1.6%

Each of the $n_m = n - d_s$ matter clusters contributes $R = n/n_m$ triangles per edge (mean-field estimate). The first-order correction to Bethe:

$$\text{correction} = \frac{R}{n(n-1)} = \frac{n/n_m}{n(n-1)} = \frac{1}{(n-1)n_m} = \frac{1}{7 \times 5} = \frac{1}{35} = 2.857\%$$

Actual gap from Bethe: 2.904%. Discrepancy: 1.6% (within simulation noise). This explains the Bethe-to-CRF gap *without free parameters*: the correction is $1/((n-1)n_m)$ where $n_m = n - d_s$ is the number of matter modes.

Two Sectors, One Coupling

Each sector has its own constant lattice:

Quantity	Space ($n = 8$)	Matter ($n_m = 5$)	Ratio
n / n_m	8	5	$8/5 \approx \varphi$
D_0	0.94003	0.90635	1.037
K^*	0.11750	0.18127	0.648
$\mathcal{G} = \alpha/D_0$	0.73737	0.76477	0.964
T_{avg}	35.57	60.02	0.593
$\beta = d_s \mathcal{G}$	2.2121	2.2943	0.964
$n(n+1)$	72	30	12/5

Approximate universality of $\mathcal{G}$

The most striking feature: $\mathcal{G}_m/\mathcal{G} = D_0/D_{0,m} = 1.037$. The universal coupling $\mathcal{G}$ varies by only 3.7% between the space sector ($n = 8$) and the matter sector ($n_m = 5$).

This near-constancy arises because $D_0(k) = k(1 - e^{-1/k})$ varies slowly with k for

moderate k . For large k : $D_0(k) \rightarrow 1 - 1/(2k)$, so $D_0/D_{0,m} \approx (1 - 1/16)/(1 - 1/10) = 15/16 \div 9/10 = 25/24 = 1.042$ (exact for the leading order).

Physical consequence: The gravitational coupling $G = \mathcal{G}c^2$ is essentially the same for all matter types, because all arise from modes with $D_0 \approx 0.92\text{--}0.94$. A single G governs all matter not by accident, but because space and matter modes share a nearly-common information scale D_0 .

Status Table

Result	Status	Precision
$n_m = n - d_s = 5$	Derived (exact)	integer exact
$n_m = 2^{d_s} - d_s$	Derived (powerset)	integer exact
$n_m = \text{non-singleton subsets of } \mathbb{R}^{d_s}$	Derived (combinatorial)	integer exact
SO(3) decomp: $1 + 3 + 1 = 5$	Derived (rep theory)	integer exact
V18 confirms 5 Psi-clusters	Confirmed	simulation
Loop correction $= 1/((n - 1)n_m)$	Hypothesis	gap 1.6%
Ratio $n(n + 1)/n_m(n_m + 1) = 12/5$	Derived (exact)	rational exact
$\mathcal{G}_m/\mathcal{G} = 1.037$	Hypothesis	3.7%
$G = \mathcal{G}c^2$ universal across matter	Consequence	$\sim 4\%$

Connection to K35

K35 states $\mathcal{G}^2/\varepsilon_0 = n(n + 1) = 72$ for the space sector. For the matter sector: $\mathcal{G}_m^2/\varepsilon_0 = 77.5 \neq n_m(n_m + 1) = 30$. K35 does not generalise directly to the matter sector, because K35 uses the measured ε_0 from the space-sector EIC dynamics.

However, the ratio $n(n + 1)/n_m(n_m + 1) = 12/5$ shows that the *structure* of K35 (triangular-number coupling) differs between space and matter by the clean factor $12/5$.

Summary

The $n = 8$ modes of CRF split as $n = d_s + n_m = 3 + 5$.

$n_m = 5$ is the number of non-singleton subsets of $\{x, y, z\}$ — the powerset com-

plement once the three spatial singletons are removed.

Under $SO(3)$: these five decompose as singlet + triplet + singlet = $\{\emptyset\} + \{\{x, y\}, \{x, z\}, \{y, z\}\} + \{\{x, y, z\}\} = \text{scalar} + \text{vector} + \text{pseudoscalar}$.

Space and matter are not two different things.

They are two ways of reading the powerset of $\mathbb{R}^3$.

Singletons become geometry.

Everything else becomes matter.